# Introduction

A registered report separates *what a study commits to* from *what the data later show*. Hypotheses, outcomes, exclusion rules, and the analysis plan are locked and time-stamped before results are observed; only then is the analysis run and interpreted. This ordering is the structural defence against the degrees-of-freedom problems that inflate false-positive rates in flexible analyses (Simmons et al. 2011) and motivates the broader preregistration and research-transparency movement (Nosek et al. 2018; Munafò et al. 2017; Chambers 2013).

This template operationalises that discipline as tested code rather than prose convention. The `registered_report` package freezes a registration payload under a content hash, validates that the required preregistration sections are present and well formed, and — after execution — compares the analysis that was actually run against the plan that was locked. Any departure must appear in a **deviation ledger** with a rationale before it can be reported, and confirmatory claims are kept mechanically distinct from exploratory ones.

To make the workflow concrete and reproducible, the template ships a **deterministic demonstration study**. It contains no real empirical data: a seeded generator synthesises a two-group dataset, and the registered label-permutation test is executed against it. Because both the data and the permutation schedule are seeded, every figure and every number in this document is byte-stable and is regenerated on demand from `scripts/generate_figures.py`. Readers forking this template should replace the demonstration fixture in `data/example_registration.json` and the synthetic data generator with their own registration and real data collection, keeping the same lock-validate-execute-compare structure.

The remainder of this report states the preregistered hypotheses (Section 3), describes the frozen methods and analysis plan (Section 4), reports the confirmatory outcome bound to the executed analysis (Section 5), records deviations (Section 6), and discusses scope and boundaries (Section 7).